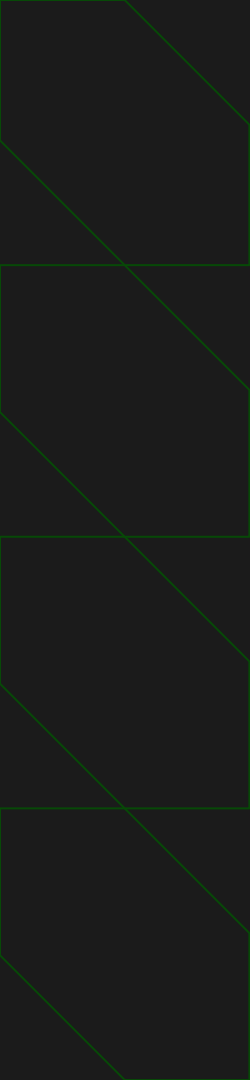


A decorative pattern of green squares and rectangles of varying shades, arranged in a grid-like fashion that tapers off towards the top right corner of the slide.

ETA Prediction

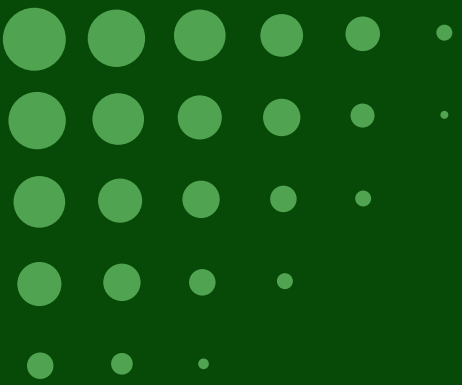
Sonia Benny Thomas
054-2503919
soniabt3@gmail.com

November 2025



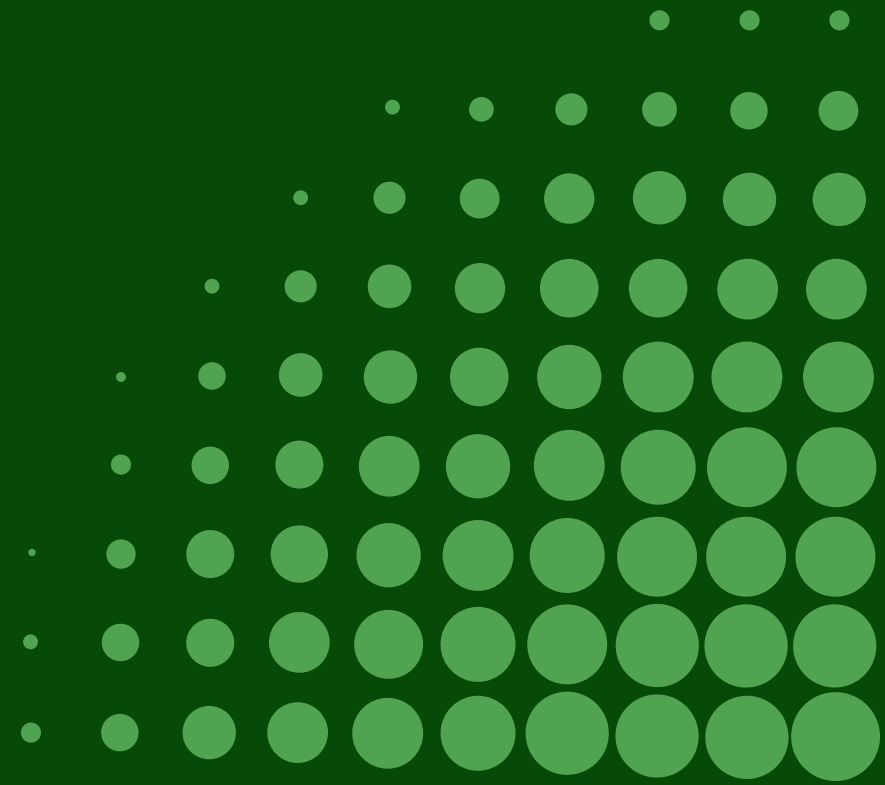
Presentation overview

- Problem Framing and Assumptions
- Synthetic Data Design and Generation
- Feature Engineering and Modelling
- Prototype



Section 1

Problem Framing and Assumptions





The problem

- Build a working version of an ETA prediction system for trips within a single city.
- Design and generate a realistic synthetic dataset and a small, runnable prototype.

Define Inputs and Outputs

Target Variable (Output) - Y

- Variable: Trip Time (denoted as trip_time_minutes).
- Description: The total duration of the trip, measured in minutes. This is the value the models (Lasso and XGBoost) are trained to predict.

Input Features - X

The models use a combination of spatial and temporal features as predictors:

Feature	Type	Description
Trip Distance	Numerical	The calculated length of the trip (in grid units).
Minute of Day	Numerical	The minute (0 to 1439) when the ride was booked Accounts for time-of-day traffic patterns.
Day of Week	Categorical	The specific day the ride occurred. Accounts for weekly trends (e.g., weekend traffic vs. weekday rush hour).
Spatial Coordinates	Numerical	The Pickup and Dropoff X and Y coordinates , which implicitly define the route and potential congestion zones.

Explain City Design

=====

CITY MAP (10x10)

=====

	(y=00)	(y=01)	(y=02)	(y=03)	(y=04)	(y=05)	(y=06)	(y=07)	(y=08)	(y=09)
(x=00)	~	~	#	.	.	#	.	.	#	.
(x=01)	~	R	#	R	.	#	.	.	#	.
(x=02)	#	#	#	#	#	#	#	#	#	#
(x=03)	.	R	#	R	.	#	.	.	#	.
(x=04)	.	.	#	.	.	#	.	P	#	.
(x=05)	#	#	#	#	#	#	#	#	#	#
(x=06)	.	.	#	.	.	#	.	P	#	.
(x=07)	.	.	#	.	.	#	.	.	#	.
(x=08)	#	#	#	#	#	#	#	#	#	#
(x=09)	.	.	#	C	.	#	.	.	#	.

Legend:

.

 = Empty

R = Residential

C = Commercial

P = Park

I = Industrial

= Road

~ = Water

Factors and Assumptions – Modelled

Category	Factor/Feature	Description	Role/Impact in Project
Modeled	Trip Coordinates	The exact starting and ending points on the simulated grid.	Define the trip's spatial geometry, which <i>implicitly</i> contains location/zone data that could be used for historical traffic pattern analysis (zonal modeling).
Modeled	Trip Distance	The calculated length of the trip (via Manhattan distance).	The fundamental variable providing the base travel time.
Modeled	Minute of Day	The time the trip was booked (0-1439).	Captures predictable intraday traffic patterns (e.g., rush hour congestion multipliers).
Modeled	Day of Week	The day the trip occurred.	Captures predictable weekly traffic trends (e.g., weekend traffic multipliers).

Factors and Assumptions – Ignored

Category	Factor/Feature	Description	Role/Impact in Project
Ignored	Route Complexity	Real-world street topology, elevation, one-ways.	Assumption: Assumes simple Manhattan Distance (grid-based movement).
Ignored	Real-Time Events	Accidents, unexpected road closures, special events.	Assumption: Modeled only via random Gaussian noise.
Ignored	Traffic Flow Dynamics	Localized congestion, cascading effects, bottlenecks.	Assumption: Assumes uniform speed across the grid for a given time window.
Ignored	Driver/Vehicle Data	Driver behavior, specific vehicle performance, and speed limits.	Assumption: Assumes uniform driver behavior and vehicle type.

'Good Enough' Baseline Model

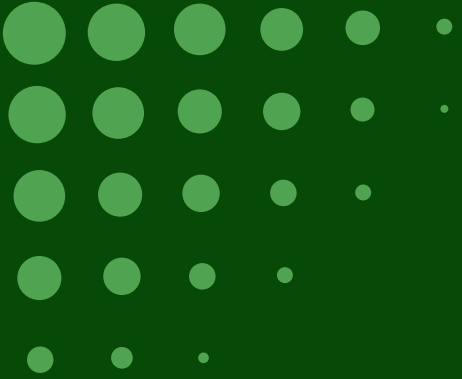
The naive model assumes constant, idealized travel conditions and uses a fundamental physics formula:

$$\text{Time} = \text{Distance} / \text{Speed}$$

- Here, we're assuming distance is the shortest manhattan distance between two points on the city map
- Speed is the average speed limit on the zones the path traverses

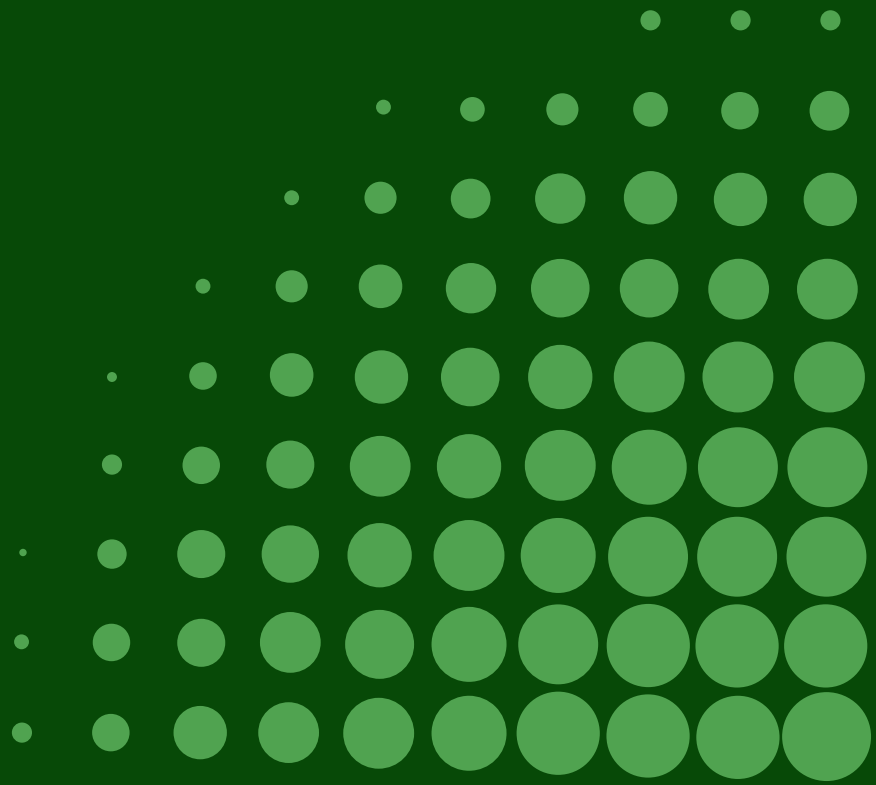
Metric	Result	Interpretation
Mean Absolute Error (MAE)	8.21 minutes	On average, the prediction is off by over eight minutes
Business Threshold (Error <= 5 min)	26.67% (4 out of 15)	The model is considered 'good' less than one-third of the time, failing to meet business requirements for accurate ETA.

Goal: The Machine Learning models must reduce the Mean Absolute Error significantly below 8.21 minutes and achieve a business success rate much higher than 26.67 %

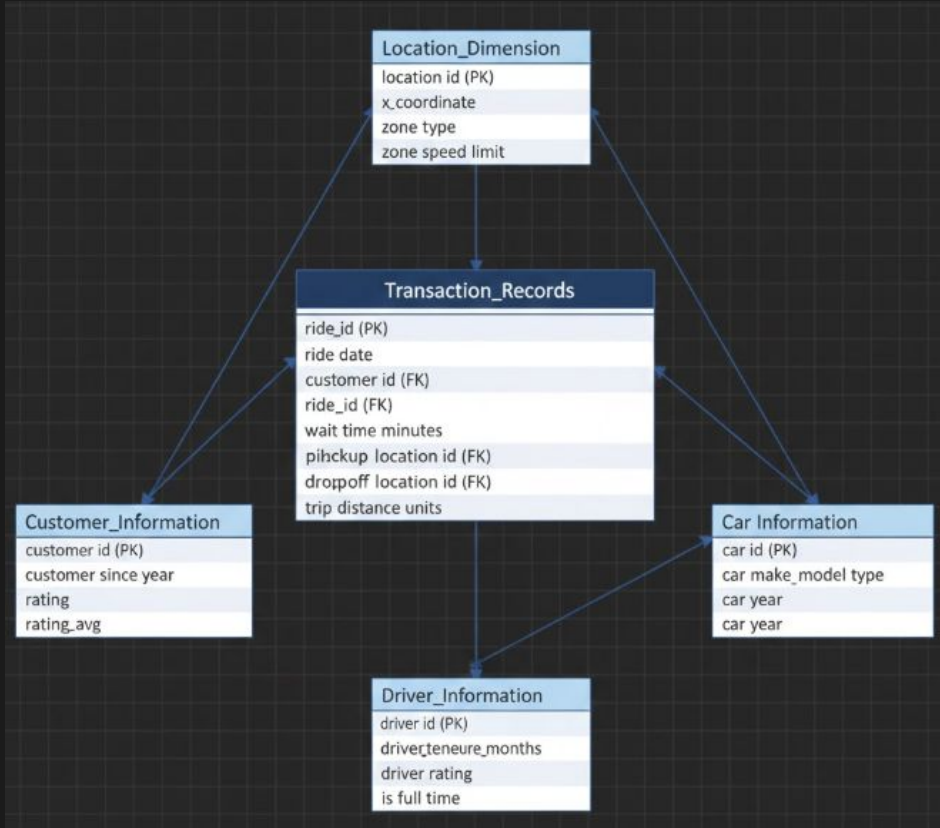


Section 2

Synthetic Data Design and Generation



Schema Design



Weighting to manage bias and relationships

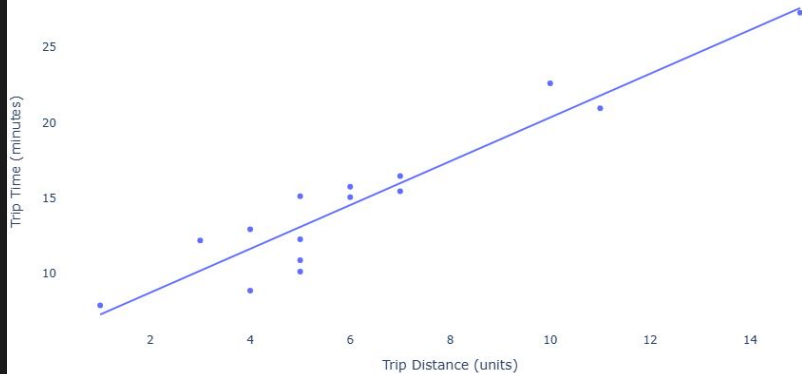
The dataset was generated using a controlled, weighted simulation to ensure the `trip_time_minutes` (target) is a function of the input features, making the data inherently predictive.

- Goal: To establish a controlled environment where the relationship between input features and the target variable is clearly measurable.
- Distance Calculation: Trip Distance between every two points is precisely calculated using the Manhattan Distance (L1 Norm) based on the four corner coordinates.
- Temporal Feature Extraction: `Day_of_week` and Minute of Day are directly extracted from the generated pickup timestamp to create clean numerical predictors.
- Base Time: The initial travel time is determined by the shortest distance / speed limit
- Time-of-Day Weight: A multiplier is applied to the base time to simulate rush hour congestion, slowing down travel during specific minute-of-day windows.
- Day-of-Week Weight: A separate multiplier is applied to simulate slower weekend traffic trends.
- Target Calculation: The trip time is the final output of the combined weighted formula, making it the dependent variable.
- Random Noise: Gaussian noise is added to the final time to simulate small, unpredictable real-world delays and prevent perfect model training.
- Outcome: This ensures that the distance, time, and day features are the *true explanatory variables* driving the model's performance.

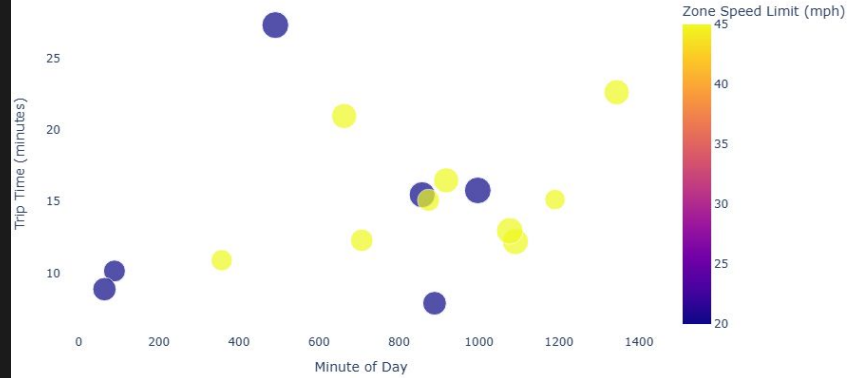
About the dataset

- Distance: The scatter plot confirms a strong, positive linear correlation, demonstrating that distance is the fundamental variable for predicting trip time, as expected.
- Weekly Trends Exist: The box plot clearly shows that median trip times and overall variance are significantly higher on weekends (Days 6 & 7)
- Rush Hours are Present: The line graph exhibits pronounced peaks in trip time, which correspond precisely to the simulated rush hour windows

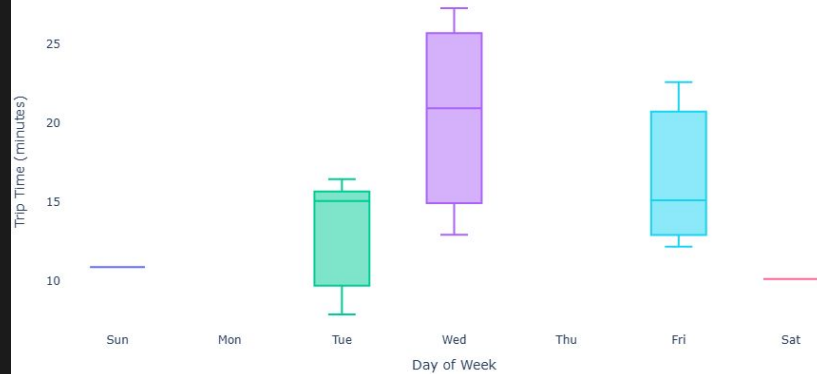
3. Relationship: Trip Distance vs. Trip Time

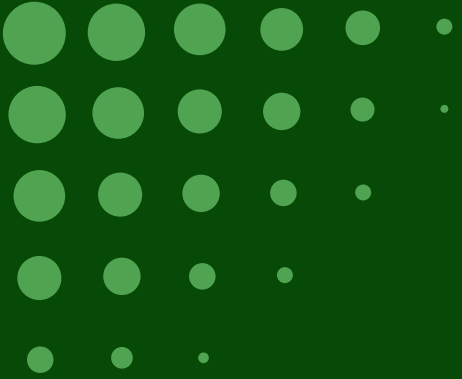


4. Temporal Effect: Trip Time by Minute of Day



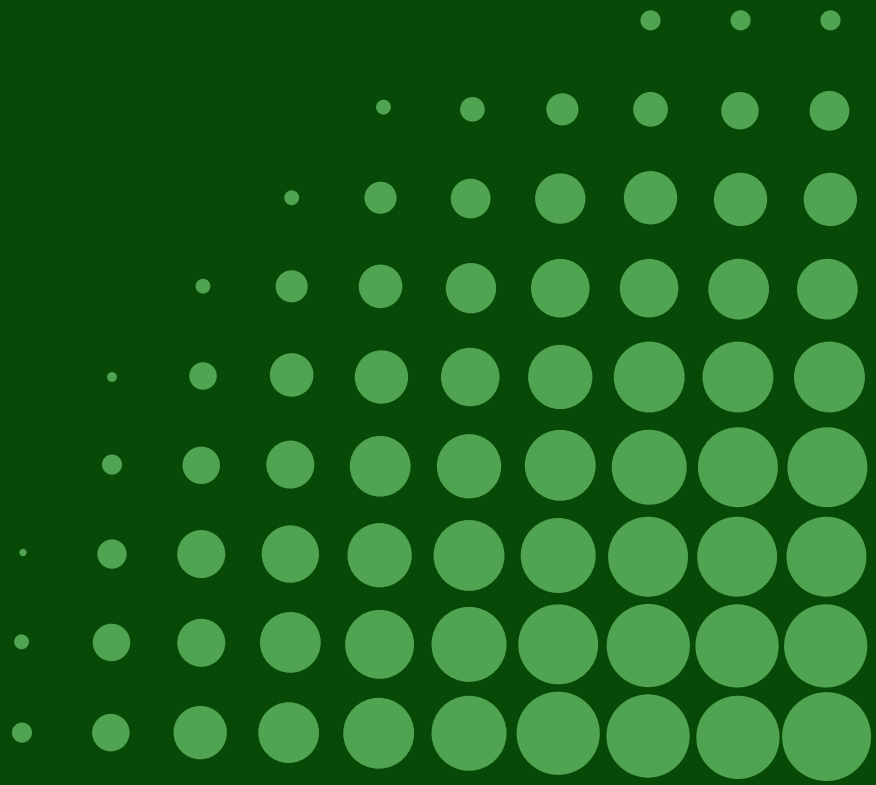
5. Categorical Effect: Trip Time Distribution by Day of Week





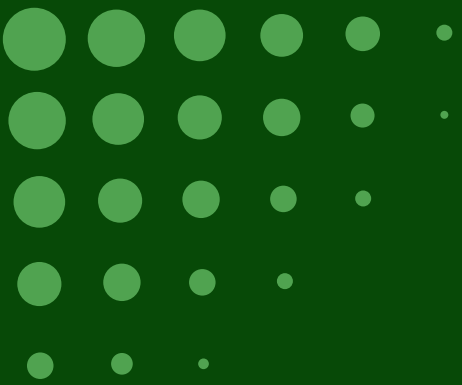
Section 3

Feature Engineering and Modelling



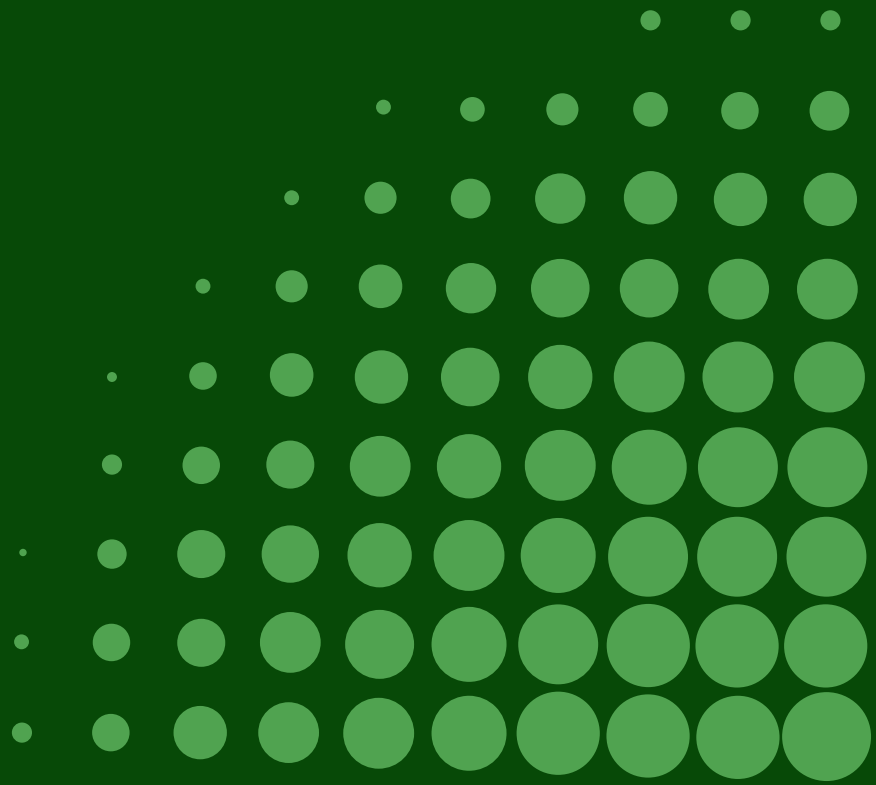
Feature Engineering and Model Performance

Feature	Naive Baseline	Lasso Regression	XGBoost Regressor
Model Type	Rule-Based / Physics Formula	Linear Model (Regularized)	Ensemble Tree-Based (Gradient Boosting)
Feature Engineering	Uses only raw Distance and Speed Limit.	Uses Distance, Minute of Day, and Day of Week.	Uses Distance, Minute of Day, and Day of Week.
Categorical Encoding	Not applicable.	Required One-Hot Encoding	Required One-Hot Encoding (OHE)
Scaling Technique	None.	StandardScaler was required to normalize numerical features due to regularization.	Scaling is generally not required for tree-based models like XGBoost.
Key Performance (MAE)	8.21 minutes	Significantly lower than Baseline. Best Performance (approx 1 min)	Approx. 2.05 mins
Final Conclusion	Insufficient for real-world ETA; establishes the performance floor.	The best predictive model, capturing prediction accuracy.	Provides a good balance of interpretability and complex non-linear feature interactions.



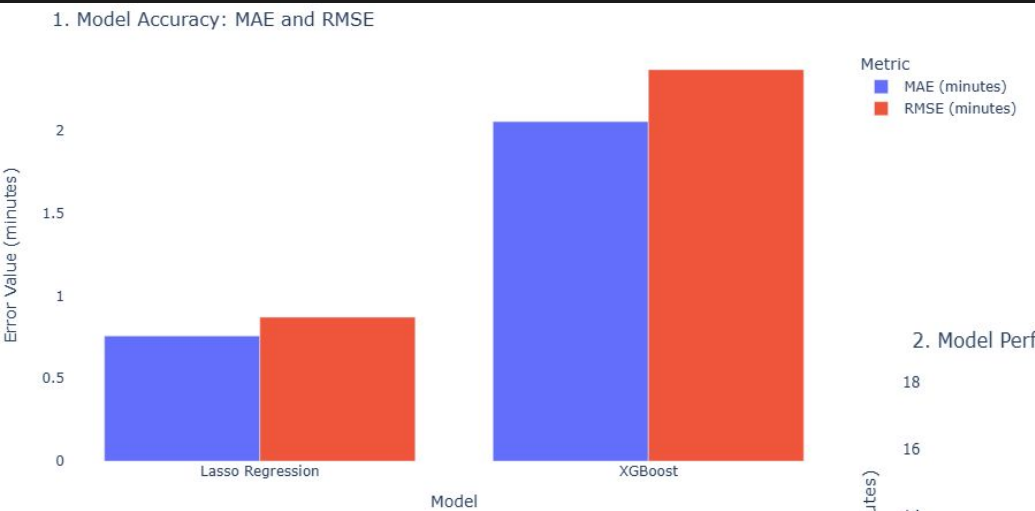
Section 4

Prototype

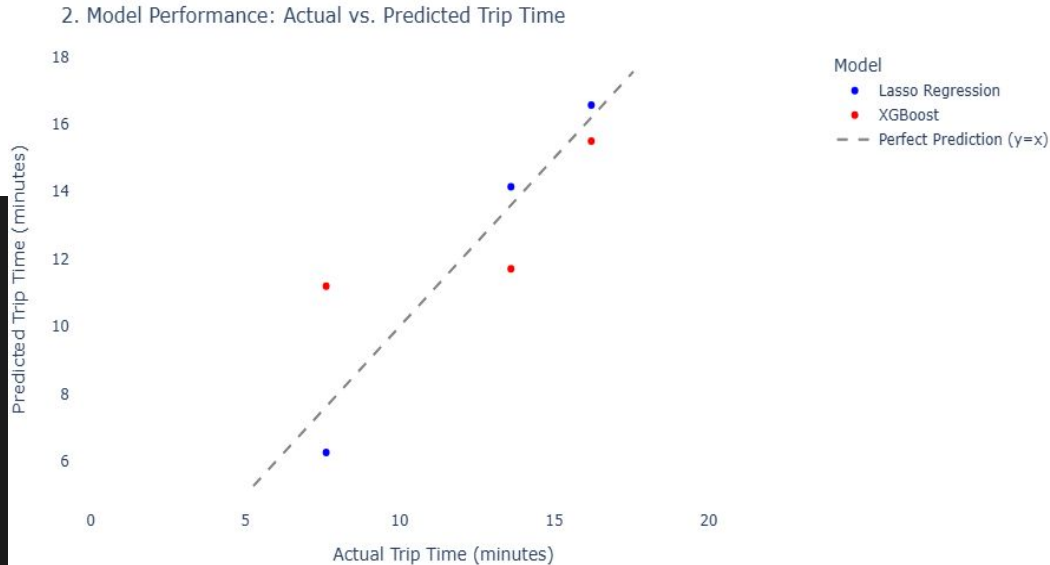


Model Comparison

1. Model Accuracy: MAE and RMSE



2. Model Performance: Actual vs. Predicted Trip Time



Dashboard

The submission file should have the following folder structure

- Code: Contains the script used for data generation, modelling and evaluation
 - **eta_prediction.py** (python file)
 - **ETA_Prediction.ipynb** (ipython notebook file)
 - requirements.txt
- Dashboard
 - **dashboard.ipynb** (uses dash and plotly to run localized dashboard)
 - **requirements.txt**
 - **model_metrics.csv** : dataset with model performance metrics, generated by ETA_predictions script file
 - **Model_predictions.csv**: test set results with predicted ETA, generated by ETA_predictions script file
 - **ride_dataset.csv**: the input data generated by ETA_predictions script file
- Documentation
 - **ETA_Prediction.pdf**: PDF version of the original ipython notebook
 - **ETA_Prediction_Presentation.pdf**: The summary presentation
 - **Case Study - Data Scientist**: Problem Statement

Steps to run code:

1. (On terminal) 'pip install -r requirements.txt'
2. (On terminal) python eta_prediction.py **or** run all cells in ipython notebook
3. To run the script for dashboard, ensure the three sql files generated by 'eta_prediction' are accessible to 'dashboard.py'